

Contribution of Neocortex to Sound Localization in Opossum (*Didelphis virginiana*)

RICHARD J. RAVIZZA AND BRUCE MASTERTON

*Psychology Department, Vanderbilt University, Nashville, Tennessee 37203; and
Psychology Department, Florida State University, Tallahassee, Florida 32306*

SINCE NEFF and his colleagues demonstrated sound localization deficits in cats with auditory cortex ablations, impairments of this sort have been shown to result from cortical lesions both in monkeys and in man himself (9, 17, 21). The apparent commonality of this role of auditory cortex among advanced mammals leads to the notion that a neocortical role in sound localization might be common to all mammals and further, might even be one of the more primitive of neocortical contributions to hearing. In order to assess these possibilities more directly, we began a series of experiments into the functions of auditory cortex in the common Virginia opossum, one of the most neurologically primitive mammals now extant (6, 18, 20).

The selection of the opossum for ablation-behavior experimentation brought with it a formidable array of problems. Wild opossums seemed to be entirely unmanageable with the usual techniques of behavioral testing—preferring to starve rather than eat in the presence of the experimenter and preferring to remain almost tetanized by electrical shock rather than move a short distance to avoid it completely. For these reasons, we were obliged to turn to a relatively new behavioral technique that had proved useful in testing other uncooperative animals and indeed, it eventually proved to be successful for opossums too. This behavioral technique makes use of an animal's natural response of stopping whatever it happens to be doing whenever a novel, unexpected, or mildly frightening situation presents itself.

In general, this technique is similar to the

sound-localization tests originally employed by Neff and his colleagues in that they also made use of an easily observable conditioned response based on a natural reaction of cats to approach the source of a sound that is associated with food. However, unlike the technique used in previous sound-localization experiments, the present one does not require the animal to move its body toward the sound source. Instead, the animal merely indicates the direction of the sound source by “freezing”—much in the same way that a person might press a button or raise his hand in response to a change in a sound source.

As it turned out, this fundamental difference in the overt response required of the animal also proved to be useful. First, it enabled strict behavioral testing both of normal and decorticate opossums. Second, it offered an opportunity to subdivide sound-localization ability into what may be two of its component parts: the perception of the direction of a sound source, on one hand, and the ability to use this information to find the sound source, on the other.

METHODS

In all, 12 wild-born adult Virginia opossums (*Didelphis virginiana*) were used. Six underwent surgery before training was begun. The other six were trained and tested without surgical intervention of any kind. It is the difference between these two groups in the outcome of the behavioral tests that form the chief results in this report.

Anatomical procedure

Although there is good agreement concerning the central focus of auditory cortex in the opossum, Fig. 1 shows that the three methods usually used to delimit this area differ somewhat

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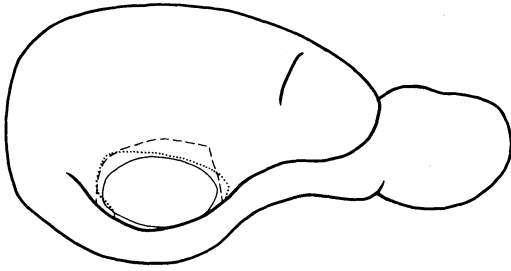


FIG. 1. Dorsolateral (45°) view of opossum forebrain adapted from Diamond and Utley (4). Auditory cortex based on cytoarchitecture is shown with dashed line (7); thalamic retrograde degeneration with solid line (2, 4); gross evoked potentials with dotted line (10).

in their conclusions concerning its extent. Furthermore, the thalamic retrograde degeneration method has shown that, in addition to the essential projection of the medial geniculate (GM) nucleus to the temporal cortex shown in the figure, there may exist sustaining projections from GM to neocortical regions well beyond temporal cortex (4). Indeed, anterograde degeneration techniques show GM fibers projecting to the peristriate-parietal junction at the medial edge of the hemisphere (5) and electrophysiological recordings from unanesthetized animals show large short-latency evoked potentials over a wide area of cortex (13).

Due to this uncertainty surrounding the limits of the area of the opossum's neocortex that might be properly considered "auditory," together with the possibility that future studies may indicate the inclusion of still more extensive areas, it seemed expedient for the purpose of these experiments to remove all of the neocortex. Therefore, the surgical procedures were aimed at complete removal of neocortex in each of six opossums.

NEOCORTICAL ABLATION. Bilateral ablation of neocortex was performed in a single-stage operation under aseptic conditions. For surgery, the animals were anesthetized with pentobarbital (30 mg/kg body wt). The coarse hair covering the skull was shaved, the scalp opened, and the temporal muscles detached from the midsagittal ridge. The skull was then opened, the dura folded to one side, and cortical tissue removed by aspiration.

After the tissue had been removed, the field was sprinkled with antiseptic powder (Neosporin) and the dura replaced. The temporal muscles were then stitched across the midsagittal ridge, and the scalp incision was closed with silk thread. Finally, the animals were given 30,000

units of penicillin and placed in their home cages for a recovery period of at least 2 weeks before behavioral testing was begun.

ELECTROPHYSIOLOGY AND HISTOLOGY. Following behavioral testing, the completeness of each lesion was verified histologically and, in three cases, electrophysiologically as well. For the electrophysiological verification, the decorticate opossums were anesthetized with 0.6 ml/kg dial-urethan and the brain exposed once more. During stimulation with a loud click, the remnants of each cerebral hemisphere were explored with silver-ball electrodes in an attempt to locate any sound-responsive cortex that might have been missed in the ablation. When these observations were completed, the animals were perfused with isotonic saline and then 10% formalin.

After perfusion, the outer and middle ears were routinely dissected and examined for evidence of malformity, damage, or disease. The brains were then removed, embedded in celloidin, and serially sectioned at 35 μ in a frontal plane. Every fourth section (140 μ) rostral to the anterior commissure and every other section (70 μ) throughout the thalamus was stained alternately by the Weil and the cresyl violet techniques. The cortical lesion was then reconstructed from systematic projection of the Weil sections and finally, retrograde degeneration in the thalamus was assessed with the cresyl violet sections.

Behavioral method

The behavioral technique employed to test the auditory capacities of the normal and decorticate opossums has been described in detail elsewhere (14, 19). Briefly, the opossums were trained to lick a water spout for an occasional reward of water. After licking became stable and persistent, a sound (or a change in sound) was presented, followed in 10 sec by an unavoidable shock to the feet. After several pairings of sound stimulus and shock, the opossums stopped licking whenever the stimulus was presented and did not resume licking until the shock had been received. In test sessions, this cessation or slowing of licking was used as evidence that the stimulus (or the change in the stimulus) had been perceived.

BEHAVIORAL EQUIPMENT. The essential features of the behavioral apparatus are shown in Fig. 2. In order to insure a maximum reduction of standing waves, background noise, echoes, and sound shadows, five features were incorporated into the design of the behavioral apparatus. First, the apparatus was placed in the center of

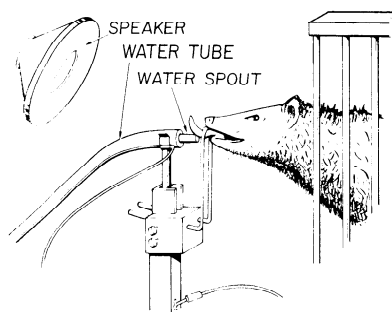


FIG. 2. Apparatus used for behavioral testing. Opossum extended its head through wall of restraining cage in order to lick water spout. By careful positioning of cage and water spout, the opossum's head could be made to remain at a relatively fixed distance from the speakers. Water spout and brass bar bent in an inverted U also served as the two sides of an electronic circuit that allowed recording of tongue contacts with water spout.

a sound-treated chamber (IAC, 1202A). Second, the platform supporting the apparatus, and the ceiling and walls of the tests chamber were loosely draped with several layers of burlap. Third, the water spout was positioned so that the animal had to extend its head and ears beyond the confines of the apparatus to reach it. Fourth, the water reservoir was connected to the spout by a long tube that allowed its placement beyond the immediate sound field. Finally, the speaker and the behavioral apparatus were isolated from each other by several layers of loose burlap and soft felt pads.

SOUND PRODUCTION EQUIPMENT. Sine waves for pure-tone stimuli were produced by an oscillator (Hewlett-Packard 200CD), and noise was produced by a white-noise generator (Grason-Stadler 455 C). The output of the generators was led first to an electronic switch (Grason-Stadler 829E), then to attenuators (Hewlett-Packard 350D), and finally, to a speaker (University 312) placed in front of the test apparatus. This sound system proved capable of delivering undistorted tones over a frequency range from 50 Hz to 80 kHz. Short-duration noise bursts for sound-localization tests were controlled with a waveform generator (Tektronix 162) and a pulse generator (Tektronix 161).

Throughout training and testing on the several discriminations, the electrical signal to the speaker was monitored with an oscilloscope which had been calibrated previously with a sound-level meter. To measure the sound pressure and to calibrate the sound system and oscil-

loscope, condensor microphones (Bruel and Kjaer 4133 or 4136) and amplifier were used. During calibration, the microphone was placed at several orientations in the general region occupied by the animal's head while it was licking. For details of the calibrating technique, see Heffner et al. (8).

The sound sources for the localization test were hearing-aid receivers (Telex 3W2) mounted on movable stands. To eliminate reverberations and vibrations, each receiver was suspended from its stand by rubber bands. Although it was not possible to obtain receivers perfectly matched for transient sounds, three pairs were carefully selected from a pool of 24 receivers on the basis of their similarity in intensity and timbre to human observers. The members of the three best-matched pairs were then randomized for each daily session. The adequacy of this matching procedure for removing possible artifacts was confirmed during regular training and testing sessions because the opossums' performance always fell to chance levels whenever the angle between the receivers became small.

BEHAVIORAL PROCEDURE. The opossums were trained to lick the spout to obtain water reward. Once they had achieved and maintained a steady rate of licking with intermittent reward (FR10), the assessment of auditory capacities was begun. Each of the operated and normal animals first underwent a series of sound-reflex tests. Since the results of these tests proved to have little or no bearing on the functions of auditory cortex, they have been reported elsewhere (15). This report consists of the outcomes of a pure-tone sensitivity test and a battery of five sound-localization tests. In order of their presentation, the tests were: 1) thresholds for pure tones, 2) thresholds for sound localization in the horizontal plane (azimuth threshold), 3) thresholds for localization in the vertical plane (elevation threshold), 4) azimuth thresholds for very brief sounds, 5) azimuth localization of sounds with low repetition rates, and 6) the identification of the side from which widely separated sounds emanate.

GENERAL TESTING PROCEDURE. Each of the six tests followed the same general pattern. A test was conducted over a series of successive daily sessions, each session consisting of 30 trials. The daily sessions themselves are most conveniently described first in abstract and then in detail.

In abstract, each session can be thought of as an hour-long series of 10-sec periods. Each of the 10-sec periods was either a "safe" ("no-

shock") period in which a safe signal was continuously presented, or a "shock-warning" period in which a "warning" signal was continuously presented. The warning periods were always begun by a shift in the signal from safe to warning. They were always terminated by a shock and a shift in the signal from warning to safe (Fig. 3). The nature of the difference in safe and warning signals, of course, depended on the particular auditory dimension that was being tested. For example, in a test of azimuth localization, the safe periods were occupied by a 1/sec train of brief noise bursts from a speaker to the right of the animals, while the warning periods were occupied by an identical train of noise bursts from a speaker to the left.

THRESHOLD DETERMINATIONS. In each of the behavioral tests that required the estimation of a threshold, the threshold was established in two ways. In the first procedure, the warning signal was systematically made more similar to the safe signal, step-by-step on each successive trial until the animal failed to respond (i.e., failed to cease licking). The warning signal was then made more unlike the safe signal on each successive trial until an obvious response occurred once more. By repeating this procedure

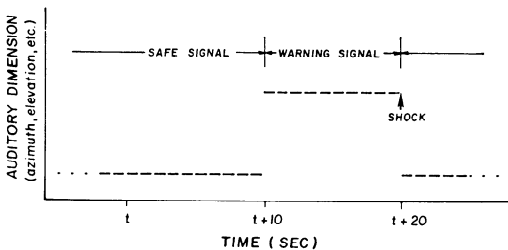


FIG. 3. Stimulus arrangement in a behavioral test trial. Time runs from left to right on the horizontal dimension. The safe signal and the warning signal are each represented by different positions along the vertical axis (i.e., for the horizontal sound-localization test the upper dashed line might be a source 30° to the left and the lower dashed line might be a source 30° to the right). The warning period consists of the 10 sec of warning signal presentation (i.e., from $t+10$ to $t+20$). The period during which the safe signal is presented is divided into many safe periods, each 10 sec in duration, but the safe period immediately preceding the warning signal (i.e., from t to $t+10$) is the safe period used for calculating differential lick rates. The arrow indicates the time at which the warning period is terminated by shock. After training, the opossum stops licking whenever the warning signal is presented and does not begin again until shock is received and safe signal is reintroduced.

a few times, the threshold was readily estimated.

In the second threshold-testing procedure, five levels of the warning signal, both above and below the previously estimated threshold, were presented in random order. After at least 10 presentations at each of these preselected levels, a smooth psychophysical function expressing the relation between the opossum's response (changes in lick rate) and the level of the warning signal could be readily plotted (14).

Test 1: pure-tone threshold tests. Threshold sensitivity for pure tones was determined at two-octave intervals (0.5, 2, 8, and 32 kHz). Unlike the localization-threshold tests, there was no sound whatever during the safe period in this test. The warning signal for the pure-tone tests consisted of a 1/sec train of tone pips of 800 msec duration with a rise and decay time of 25 msec for 2, 8, and 32 kHz, and 50 msec for 0.5 kHz. For threshold determination, the intensity of the tones was varied in 6-dB steps.

Test 2: horizontal- (azimuth) localization threshold. In the second test, the acuity for the azimuth of a sound source was measured. The safe signal consisted of a 1/sec train of white-noise bursts (100 msec duration, 50 dB SPL, fast onset time) 15 inches to the right of the animal. The warning signal was a similar train of noise bursts, 15 inches to the left of the animal.

During the first stage of testing the angular disparity of the two sound sources was systematically varied until threshold estimates could be made. Since these initial tests showed the localization threshold for both normal and decorticate opossums to be relatively small, a quasilogarithmic series of angular separations (180° , 120° , 60° , 30° , 15° , 7.5° , 3.75° , and 0°) was selected for the second stage of testing (Fig. 4).

In the second stage of threshold testing, a daily session consisted of trials at each of three angular separations. Each angle was represented by a pair of matched receivers. Throughout any one daily test session, the smallest angle of separation was always near or below the previously estimated threshold, whereas the other two angular separations were at or above the estimated threshold.

Tests 3, 4, 5, and 6. In an effort to explore the limits of sound-localization ability in the decorticate opossums more fully, four further localization tests were presented. However, since each of these tests was only one or another variation of the horizontal localization test already described, their description and the reasoning behind their design can best be presented as the results of the earlier tests unfold.

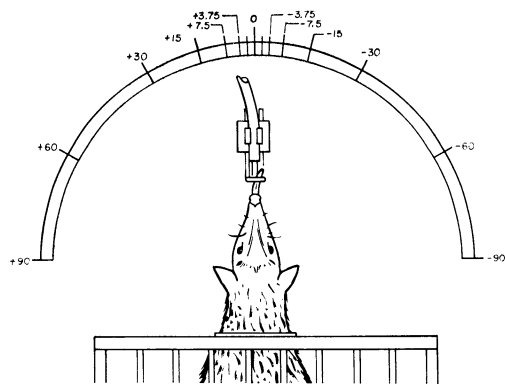


FIG. 4. Top view of test apparatus for horizontal sound-direction tests. For a given angular disparity, say 120°, the safe signal was a train of brief noise bursts from an earphone located at -60°. For the warning signal, the train of noise bursts was shifted to a second earphone at +60°. The sounds from +60° continued for 10 sec until a shock terminated the trial. The train of sounds was then shifted back to the earphone at -60°.

RESULTS

Anatomical results

The anatomical results are summarized in Table 1. The reconstruction of the corti-

cal lesion and drawings of representative sections through the thalamus of the case with the smallest lesion (opossum 19) are shown in Figs. 5 and 6.

Table 1 shows that there was little variation either in the extensiveness of the cortical lesion or in the severity or extent of the resulting retrograde degeneration in the thalamus. In every one of the six cases, retrograde degeneration in the thalamic auditory nuclei (GM and Po), though incomplete in the sense that a few scattered neurons remained, proved to be as severe as the degeneration that results from a complete decortication (4).

The only variation among the six cases was in the degree of degeneration in the anterior (AV, AD, and AM) and medial dorsal (MD) nuclei of the thalamus. In two cases, (opossum 9 and 43) the medial wall of the cortex was completely undercut by the lesion and consequently, the entire neocortex was either removed or disconnected from the thalamus in these cases. In the remaining four cases (opossum 19, 28, 42, and 44), medial neocortex was neither completely removed nor completely undercut.

TABLE 1. Anatomical results

Animal number	9	19	28	42	43	44
Evoked responses to sound	-	-	None	-	None	None
Cortical ablation						
Auditory cortex	C	C	C	C	C	C
Lateral neocortex	C	I*	C	C	C	I*
Medial neocortex	I'	I	I	I	I'	I
Retrograde degeneration						
GM	M	M	M	M	M	M
Po	M	M	M	M	M	M
GL	M	M	M	M	M	M
LP	M	M	M	M	M	M
V	M	M	M	M	M	M
L	M	M	M	M	M	M
AD	M	P	P	P	M	M
AV	M	P	P	P	M	M
AM	M	M	M	M	M	P
MD	M	M	M	M	M	P
SPT	M	M	M	M	M	M
Conclusion						
Auditory cortex	C	C	C	C	C	C
Neocortex	C	I	I	I	C	I

- = subject died before recordings could be made; C = complete removal; I = incomplete removal; ' = undercut; * = anterior portion of lateral wall, no insular or temporal cortex; M = maximal degeneration; P = probably less than maximal degeneration. See Fig. 6 for abbreviations of thalamic nuclei.

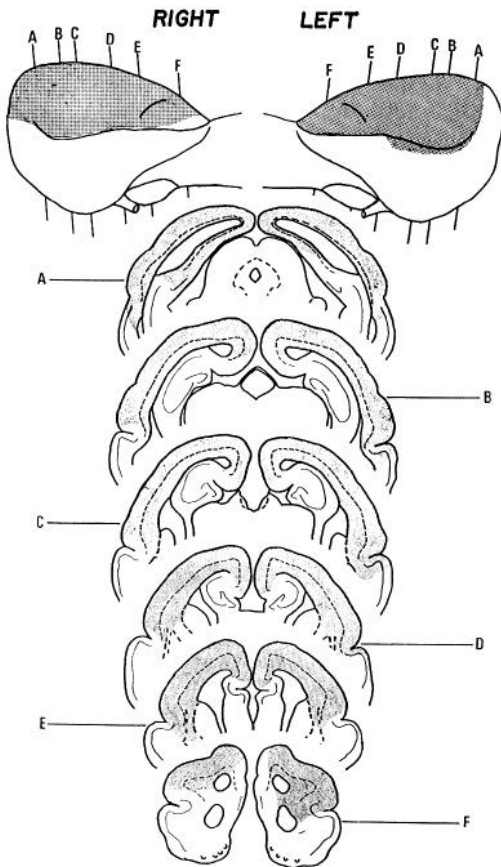


FIG. 5. Reconstruction of the cortical lesion in the case with the smallest lesion (opossum 19). Of the neocortex that escaped direct ablation, only the tissue along the medial wall might have remained connected to the thalamus (section F). Subcortical damage was limited to a small nick in the right caudate and another in the left putamen (section D).

In these cases, small regions of the anterior or medial nuclei were somewhat less than maximally degenerated.

Finally, in no case was there any sign of disease, damage, or malformation of either the outer or middle ears, though in one case (opossum 28), there was an excessive accumulation of wax in the external auditory meatus.

In summary, each of the six decorticate opossums suffered complete removal of auditory cortex whether that term is defined cytoarchitectonically or by the resulting retrograde degeneration in the thalamus. Furthermore, in those cases examined elec-

trophysiologically, exploration of the perimeter of the lesion revealed no auditory evoked responses of neocortical origin.

Behavioral results

Before describing the outcomes of the behavioral tests of sound localization ability, it is important to note that the opossums had been tested previously for their absolute sensitivity to white noise (15). For the purposes of the present report, it need only be noted that each of the six decorticate opossums was virtually identical to normal opossums. Therefore, the possibility that a deficit in localizing noise bursts might be a secondary result of total or partial deafness can be ruled out. Nevertheless, it also seemed advisable to rule out the possibility of a hearing loss restricted to one or another portion of the audible field. Therefore, thresholds for pure tones were also obtained before proceeding with the localization tests.

TEST I: PURE-TONE SENSITIVITY. The results of the pure-tone threshold tests are shown in Fig. 7. It can be seen that the sensitivity of a decorticate opossum to pure tones differs little from normal opossums over a wide range of frequencies. Only at the very lowest limits of the audible field might there be some question of abnormality and on postmortem examination, this animal was found to have an excessive amount of wax in its outer ear. Therefore, the slight deviation from normal at 500 Hz does not seem great enough to warrant speculation.

When the results of this test are combined with the results of the previous white-noise sensitivity test, it can be concluded that the removal of auditory cortex results in little or no change in an opossum's audible field. Certainly, decortication produces no change large enough to suggest its use as an explanation of other deficits.

TEST II: HORIZONTAL (AZIMUTH) SOUND-LOCALIZATION ACUITY. Turning now to the sound-localization tests themselves, Fig. 8 shows the results of the azimuth-threshold tests. Before detailing the difference between normal and decorticate opossums, however, it should be noted that both normal and de-

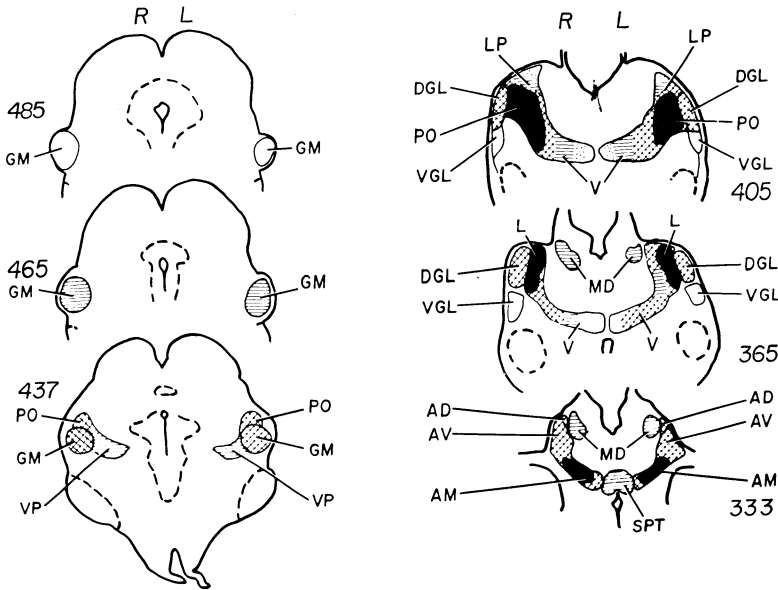


FIG. 6. Thalamic degeneration resulting from the lesion shown in Fig. 5 (opossum 19). Black: no neurons remain; crosshatch; some neurons remain; stripe: many neurons remain; stipple: some degenerative changes present. AD, n. anterior dorsal; AM, n. anterior medial; AV, n. anterior ventral; DGL, dorsal lateral geniculate body; GM, medial geniculate body; L, n. lateral; LP, n. lateral posterior; PO, n. posterior; VGL, ventral lateral geniculate body; VP, n. ventral posterior. Except for AD and AV, thalamic nuclei are degenerated to the same degree that results from perfect decortication. Note complete preservation of caudal GM.

corticate opossums were able to detect large shifts in the direction of the sound source, but not small shifts. This one observation serves to rule out two possible explanations of the results in this and the following localization tests. First, the inability of the opossums to detect small changes in azimuth demonstrates that the high level of discrimination at wide angles is due to a true sensitivity to the direction of the sound source and not to an ability to discriminate slight qualitative differences that might have existed between the left and right sounds themselves. Therefore, the conclusion that opossums can detect a change in the direction of a sound source, with or without benefit of auditory cortex, seems inescapable. Second, the ability of decorticate opossums to respond to large shifts in direction shows that any deficit that might appear in this or later tests is not due to a more general inability to meet the strictly motor, emotional, or intellectual requirements of the behavioral tests.

Returning to the results in Fig. 8, it can be seen that even though decorticate opos-

sums can detect a horizontal shift in the direction of a sound source, they are far from normal: the mean threshold of the normal opossums was 4.6° while the mean threshold of the decorticate opossums was 24.1° , a fivefold increase (Fig. 9). Therefore, it can be concluded that decorticate opos-

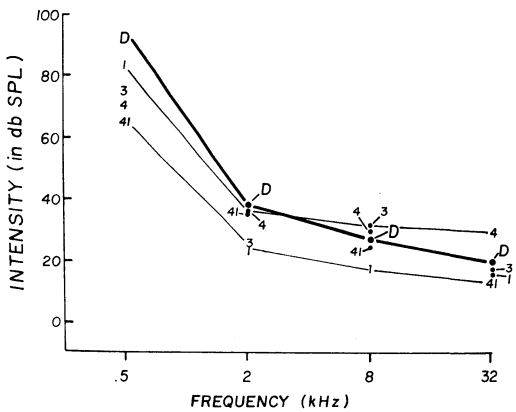


FIG. 7. Results of pure-tone threshold tests for four normal and one decorticate (D) opossum (case 28). Decortication has little effect on audible field.

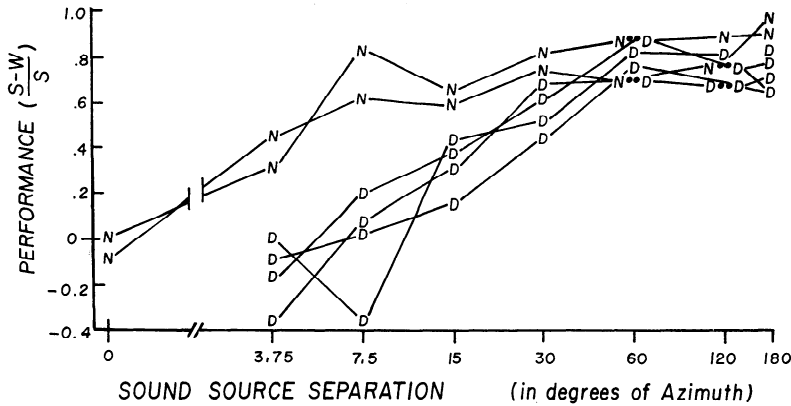


FIG. 8. Performance of four decorticate (D) and two normal (N) opossums on horizontal sound-direction test. Behavioral performance is in units of change in response rate. S = number of responses during safe signal (sound on right), W = number of responses during warning signal (sound on left). Performance axis is scaled so that perfect detection of warning signal is represented by 1.0 and no detection by 0.0. Note logarithmic scaling of abscissa.

sums suffer a marked decrease in acuity for a shift in the azimuth of a sound source.

Finally, Fig. 9 shows that the difference in directional acuity in normal and decorticate opossums is independent of the definition of threshold. The parallel psychophysical functions for normal and decorticate opossums imply that the decorticates were as vigilant as normals in their efforts to detect the shift in direction. This observation suggests that the acuity shift resulting from decortication is not merely a secondary result of a loss of alertness.

TEST III: VERTICAL (ELEVATION) SOUND-LOCALIZATION ACUITY. Since the integration of binaural cues plays a crucial role in the horizontal localization of sound sources but not in vertical localization, the partial success of the decorticate opossums in detecting a shift in azimuth yields no clue as to whether or not they can detect a shift in elevation (3). Therefore, the third test assessed the opossum's ability to detect a vertical shift in direction.

The results of this test for one decorticate and one normal opossum are shown in Fig. 10. The figure shows that decorticate opossums can indeed detect a shift in the elevation of a sound source and further, are not much different from normal.

In summary, the results of the first two sound-direction tests show that decorticate

opossums can detect a shift in either the azimuth or elevation of a sound source. Only a lowered acuity for a shift in azimuth seems to distinguish decorticate from normal opossums.

TESTS IV AND V: INTENSITY-SCANNING TESTS. Before it can be concluded that decorticate opossums are still able to localize sounds, in the usual sense of the term, it is necessary to rule out the possibility that they perform the discrimination by scanning the sound field for the direction of maximum sound intensity. For example, Riss (16) has shown that cats without auditory cortex are eventually able to locate the source of continu-

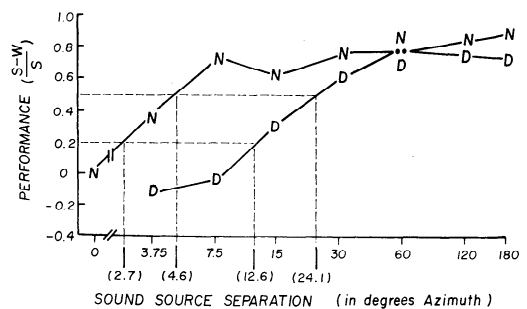


FIG. 9. Average performance of normal (N) and decorticate (D) opossums on horizontal sound-direction test. Computations of threshold values are shown using both 0.5 and 0.2 performance criteria. Note that acuity loss does not depend on the arbitrary definition of threshold acuity nor on a decrement in asymptotic performance.

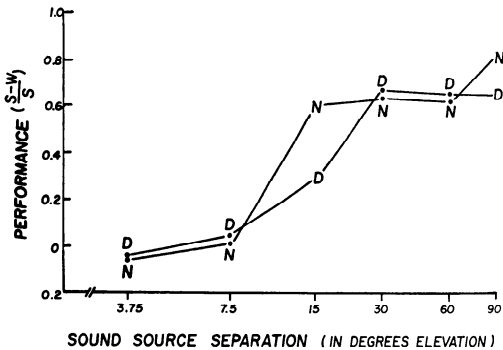


FIG. 10. Performance of normal (N) and decorticate (D) opossum on vertical sound-direction test. Note similarity in performance. See legend of Fig. 8 for explanation of performance axis.

ous sounds (such as water running from a faucet) merely by integrating intensity cues with scanning or tracking movements. Therefore, it was important to determine whether the decorticate opossums were capable of discriminating the direction of the sound sources without benefit of intensity scanning.

To begin with, three indirect observations suggested that the decorticate opossums had not relied on scanning movements in the previous tests. First, the apparatus disallowed tracking movements and large scanning movements because its design required the opossums to maintain their heads and bodies in approximately the same position in order to reach the water spout with their tongues. Second, careful observation during the previous tests revealed no systematic head movements concurrent with the warning signal. Finally, throughout the previous sound-localization tests, the opossums usually stopped licking immediately on the onset of the warning stimulus (i.e., before enough time had elapsed for the sound field to be scanned).

Despite these reasons to believe the opossums were not scanning the sound field, two specific tests were conducted to determine if intensity scanning had played any role whatever in the previous tests of directional acuity. The first (test IV) reduced the information obtainable through intensity scanning by shortening the duration of the individual noise pulses within the train.

Since a very brief sound does not allow sufficient time for a scanning movement to occur, an animal that is localizing a sound source by scanning would be expected to suffer a decrement in performance as the duration of the individual noise bursts is reduced. Accordingly, the duration of the noise bursts was reduced from 100 msec to either 10 msec or 1 msec, and the azimuth-acuity test was repeated. The results of this test are shown in Fig. 11.

It is evident from the figure that even with extremely brief sounds, decorticate opossums are still able to detect a shift in the azimuth of a sound source. Therefore, scanning movements that might have occurred during the individual sounds within the train are not the explanation of a decorticate opossum's ability to detect a shift in azimuth. However, the possibility of intensity scanning was not completely eliminated by this test.

The previous test served to rule out the possibility that cues to locus depended on scanning movements during the duration of any one pulse, but it did not preclude the possibility that the entire train of pulses was being scanned. Therefore, in a second test (test V), one of the decorticate opossums was required to detect a shift in direction of a train of noise bursts (10 msec duration) presented at a rate of one noise burst every 1, 2, or 3 sec.

The results of this test indicate that the decorticate opossum suffers only a slight de-

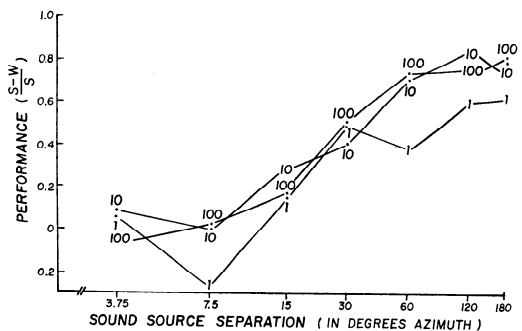


FIG. 11. Performance of decorticate opossum on horizontal sound-direction test with brief noise bursts. 100: 100 msec duration noise burst; 10: 10 msec duration; 1: 1 msec duration. Reducing the duration of the noise bursts may affect asymptotic performance, but not detection threshold.

crement in discrimination when the pulse rate is reduced to one-third the usual rate (Fig. 12). Therefore, it can be concluded that long-term scanning (i.e., over a period of several seconds) is also not the basis for the ability of a decorticate opossum to detect a shift in the azimuth of a sound source.

The results of the two tests, taken together, imply that the decorticate opossum can localize a sound source without scanning the sound field for differences in intensity. Therefore, it can be concluded that despite a loss in acuity, the cues used by decorticate opossums to detect a shift in the azimuth of a sound source are probably binaural and thus, not obviously or grossly different from the cues used by normal opossums.

TEST VI: IDENTIFICATION OF DIRECTION OF A SOUND SOURCE. The final test in the battery was designed to determine whether the decorticate animals were capable of anything more subtle than the detection of a change in the azimuth of a sound source. This test was based on the difference between discriminations of detection and discriminations of identification (Fig. 13) (e.g., ref 1). The difference between identification of the direction of a sound source and the detection of a change in direction is somewhat analogous to the difference between the capacity for "perfect pitch" and the capacity for "relative pitch" in frequency discrimination. Because the previous tests required only the detection of a change in the direction of a sound source, they are

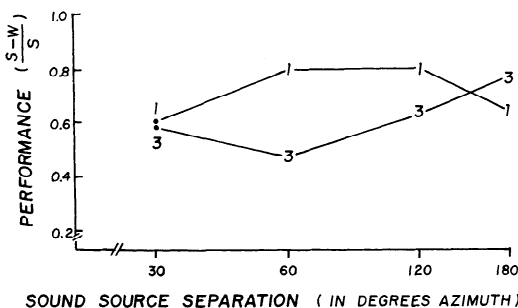


FIG. 12. Performance of decorticate opossum on horizontal sound-localization test with low noise-repetition rates. 1: 1 noise burst/sec; 3: 1 noise burst/3 sec. Lower repetition rate has negligible effect on performance.

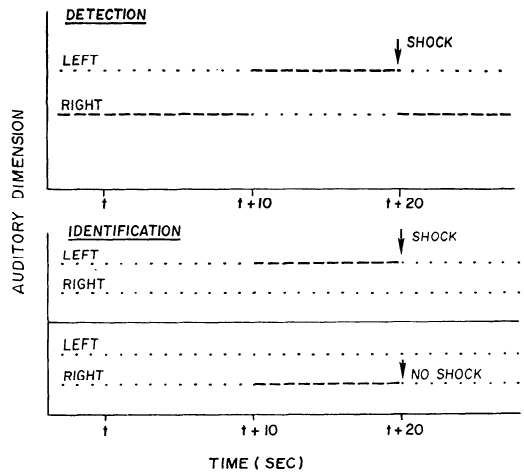


FIG. 13. Stimulus-shock relationships in tests for detection (above) and identification (below). Time from left to right on horizontal axis. For the detection tests, the background stimulation is a train of sounds from the right (safe signal) while the warning signal is an identical train of sounds from the left. For the identification test, the background is complete silence and there are two signals: either a train of sounds from the right (not followed by shock), or a train of sounds from the left (followed by shock).

analogous to a test of relative pitch. This final test was analogous to a test of perfect pitch. It was an attempt to answer the question: can a decorticate opossum identify the direction of a sound source as "on the right," or "on the left," without the aid of a second sound source for reference?

The results of the identification test can be seen in the behavioral records shown in Fig. 14. Beginning in the first session, and certainly by the end of the fourth block of trials (i.e., 80 trials), the decorticate opossums succeeded in correctly identifying the direction of the sound source without benefit of a reference. From the results of this test, it can be concluded further that decorticate opossums are not only capable of detecting a change in the direction of a sound source (if the change is large enough) but they are also capable of correctly identifying the side on which the sound source is located.

DISCUSSION

The results show that opossums deprived of neocortex suffer a sharply defined loss of

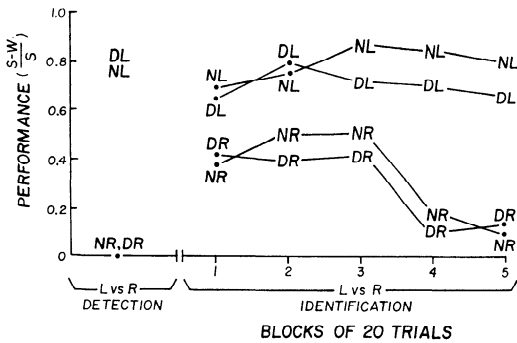


FIG. 14. Behavioral record of normal (N) and decorticate (D) opossums on detection and identification tests. L, response rate to sounds from left; R, response rate to sounds from right. Decorticate opossums can identify the side of the sound source as well as normal opossums.

directional acuity for the azimuth of a sound source. In contrast, little if any impairment exists in their sensitivity to sound, in their accuracy in detecting a shift in the elevation of a sound source, or in their ability to identify the direction of widely separated sound sources. Therefore, the discussion can be conveniently limited to an interpretation of the directional acuity test results.

To begin with, an impairment in sound localization resulting from ablation of neocortex is consistent with conclusions derived from previous sound localization and sound lateralization experiments in other mammals (cf. ref 11).

Furthermore, since the opossum's brain is one of the closest approximations to the neurological configurations that existed in the ancient common ancestry of mammals, the dependence of sound localization on auditory cortex in the opossum, specifically, has still wider implications (6, 20). First, the neurological primitiveness of opossums adds strength to the idea that the involvement of neocortex in sound localization may be common to all modern viviparous mammals because it is these mammals that count creatures with opossumlike brains among their early mammalian ancestry. Second, the dependence of accurate sound localization on neocortex in the opossum suggests that this role may be one of the oldest and most persistent contributions of neocortex to hearing. When this possibility

is combined with the obvious biological utility of accurate sound localization, it is difficult to avoid the notion that selective pressure on sound-localization ability might have been one of the primary causal factors for the early involvement of mammalian neocortex in hearing.

Although a contribution of mammalian neocortex to sound localization now seems more likely to be a general rule than an exception, it should be noted that the impairment resulting from neocortex ablation may be of different degrees in different mammals. Since this variation in deficits might provide one inroad to understanding the nature of the contributions of neocortex, the remainder of the discussion is directed to this more subtle issue.

Nature of deficit resulting from auditory cortex ablation in opossums and cats

Neff and his colleagues (11) and Riss (16) have shown that cats with large bilateral lesions of auditory cortex are completely unable to find the locus of the source of a brief sound. In apparent contrast, we have shown that completely decorticate opossums suffer a loss of directional acuity but retain an ability to identify the direction of widely separated brief sounds. At first glance, therefore, it might appear that without auditory cortex, cats suffer a more profound deficit than opossums without neocortex. However, before accepting the conclusion that there is an inherent difference in the contribution of auditory cortex to sound localization in opossums and cats, the potentially important differences in the behavioral techniques that are used for measuring sound localization abilities in the two animals must be considered. The implications of these differences in behavioral technique can be seen by examining the requirements of the tests used for cats as opposed to the requirements placed on the opossum in the experiments reported here.

In the experiments on cats, a sound was presented at one or another location and the animals were required to find the locus of its source. As Neff and co-workers (12) point out, this requirement means that a

cat's successful performance on the test is a joint function of several components. For example, the animal has to be able 1) to determine the azimuth of the sound source (a strictly sensory requirement), 2) to move the body in a given direction (a strictly motor requirement), 3) to associate the two previous capacities in order to move the body in the direction of the sound (what might be called a sensorimotor requirement)—just to name the most obvious. Clearly, using this behavioral technique, a deficit in any one of these abilities would result in a failure. However, by showing that a variation of the stimulus, either from sound to light (12) or from one sound to another sound (16), resulted in a marked improvement in performance, these same experimenters demonstrated that the complete failure of cats to localize the source of a brief sound was not due to an inability to localize the source of any stimulus—only some. Consequently, all of the possible component deficits could be ruled out except two: either the cats were unable to perceive the azimuth of the sound source (a perceptual deficit) or they were unable to act appropriately on their perception. This second possibility might be thought of as a sensorimotor deficit, such as a disruption in the functional link between auditory azimuth and motor azimuth.

Turning now to the tests in the present study, the essential difference in requirements becomes clear. The technique for testing the opossums placed no requirement whatever on the response that the animal was to make. In order to signal the direction of the sound source, the opossum did not have to move at all, and certainly not in one or the other direction. In order for the opossums to be scored "correct," however, the direction of the sound source had to be identified and if the source were on the left, the response of licking had to be inhibited. This difference in the nature of the required response means that, for opossums, an auditory azimuth-motor azimuth isomorphism was not necessary for successful performance because there was no azimuth dimension on the motor response to begin with. Therefore, the sensorimotor requirements placed on the opossums would seem

to be much less stringent than those placed on the cats. The cats had to align the azimuth of their response with the azimuth of the sound source, while the opossums did not.

Therefore, the possibility arises that opossums and cats without auditory cortex do not suffer different deficits at all. Instead, both might have an acuity impairment and an auditory-motor impairment. If this were the case, the acuity deficit was not seen in the cat experiments because the sensorimotor deficit masked it and the sensorimotor deficit was not seen in the opossum experiments because the tests bypassed it.

Whether or not this interpretation or others similar to it eventually proves to be correct, we would hesitate to cite the present report as providing crucial evidence either for or against the idea that the deficit resulting from complete auditory cortex ablation might be different for opossums than it is for cats. For the present it would seem to be more useful to reserve judgement on this issue and instead, emphasize that one inroad to understanding the contribution of neocortex to hearing might lie in the analysis of the response requirements of the behavioral procedures.

S U M M A R Y

Normal and decorticate opossums were trained and tested on a battery of auditory discriminations requiring the detection of the direction of a sound source. The results show that decorticate opossums can identify the direction of a sound source but suffer a sharply defined loss in azimuth acuity. The results also show that the loss in acuity is not merely a secondary effect of a change in the audible field, a loss in alertness, nor due to a more general deficit in either motor, emotional, or intellectual prowess.

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R. J. Ravizza was a Public Health Service Pre-doctoral Research Fellow. His present address: Psychology Dept., Duke University, Durham, N.C. 27706.

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